Progress in Photovoltaics: Research and Applications

WILEY

PROGRESS IN PHOTOVOLTAICS

SHORT COMMUNICATION

Solar Cell Efficiency Tables (Version 67)

Martin A. Green[1] | Ewan D. Dunlop[2] | Masahiro Yoshita[3] | Nikos Kopidakis[4] | Karsten Bothe[5] | Gerald Siefer[6] | Xiaojing Hao[1] | Jessica Yajie Jiang[1]

[1]Australian Centre for Advanced Photovoltaics, School of Photovoltaic and Renewable Energy Engineering, University of New South Wales, Sydney, Australia | [2]European Commission – Joint Research Centre, Ispra, Varese, Italy | [3]Renewable Energy Research Center (RENRC), National Institute of Advanced Industrial Science and Technology (AIST), Tsukuba, Ibaraki, Japan | [4]National Renewable Energy Laboratory, Golden, Colorado, USA | [5]Calibration and Test Center (CalTeC), Solar Cells Laboratory, Institut für Solarenergieforschung GmbH (ISFH), Emmerthal, Germany | [6]Fraunhofer-Institute for Solar Energy Systems – ISE CalLab, Freiburg, Germany

Correspondence: Martin A. Green (m.green@unsw.edu.au)

Received: 30 September 2025 | **Revised:** 6 December 2025 | **Accepted:** 26 December 2025

Keywords: energy conversion efficiency | photovoltaic efficiency | solar cell efficiency

ABSTRACT

Consolidated tables showing an extensive listing of the highest independently confirmed efficiencies for solar cells and modules are presented. Guidelines for inclusion of results into these tables are outlined and new entries since July 2025 are reviewed.

1 | Introduction

Since January 1993, *Progress in Photovoltaics* has published six monthly listings of the highest confirmed efficiencies for a range of photovoltaic cell and module technologies [1–3]. By providing guidelines for inclusion of results into these tables, this not only provides an authoritative summary of the current state-of-the-art but also encourages researchers to seek independent confirmation of results and to report results on a standardized basis. In Version 33 of these Tables [4], results were updated to the new internationally accepted reference spectrum (International Electrotechnical Commission IEC 60904-3, Ed. 2, 2008).

The most important criterion for inclusion of results into the Tables is that they must have been independently measured by a recognized test center listed in Versions 61 and 62 (also updated in 64). A distinction is made between three different eligible definitions of cell area: total area, aperture area and designated illumination area, as also defined elsewhere [2] (note that, if masking is used, masks must have a simple aperture geometry, such as square, rectangular or circular—masks with multiple openings are not eligible). "Active area" efficiencies are not included. There are also certain minimum values of the area sought for the different device types (above 0.05 cm^2 for a concentrator cell or one-sun "notable exception," 1 cm^2 for a one-sun cell, 10 cm^2 for a "minimodule," 200 cm^2 for a "submodule," and 800 cm^2 for a module).

Tabled results are reported for cells and modules made from different semiconductors and for sub-categories within each semiconductor grouping. From Version 36 onwards, spectral response information is included (when possible) in the form of a plot of the external quantum efficiency (EQE) versus wavelength, either as absolute values or normalized to the peak measured value. Current–voltage (IV) curves have also been included where possible from Version 38 onwards.

Three notable changes were introduced in Version 66. For large, commercially sized one-sun silicon cells, as well as the best outright result, the best result measured on a "total area" basis (only) will be included for the different competing commercial technologies as "notable exceptions." Also, the minimum area acceptable for other one-sun "notable exceptions" is 0.05 cm^2. Finally, an additional table was introduced for high bandgap

https://doi.org/10.1002/pip.70068

TABLE 1 | Confirmed single-junction terrestrial cell and submodule efficiencies measured under the global AM1.5 spectrum (1000 W/m^2) at 25°C (IEC 60904-3: 2008 or ASTM G-173-03 global).

Classification	Efficiency (%)	Area (cm^2)	V_{oc} (V)	J_{sc} (mA/cm^2)	Fill factor (%)	Test center (date)	Description
Silicon							
Si (crystalline cell)	**27.7 ± 0.5**[a]	**175.2 (t)**	**0.7458**	**42.83**[b]	**86.7**	**ISFH (10/25)**	**LONGI, n-type HTBC** [5]
Si (crystalline cell)	**27.9 ± 0.5**[a]	**158.66 (da)**	**0.7447**	**43.01**[b]	**87.1**	**ISFH (10/25)**	**LONGI, n-type HTBC** [5]
Si (thin-film minimodule)	10.5 ± 0.3	94.0 (ap)	0.492[c]	29.7[c,d]	72.1	FhG-ISE (8/07)	CSG Solar (< 2 μm on glass) [6]
III–V cells							
GaAs (thin-film cell)	29.1 ± 0.6	0.998 (ap)	1.1272	29.78[e]	86.7	FhG-ISE (10/18)	Alta Devices [7]
InP (crystalline cell)	24.2 ± 0.5[f]	1.008 (ap)	0.939	31.15[g]	82.6	NREL (3/13)	NREL [8]
Thin film chalcogenide							
CIGS (cell) (Cd-free)	23.35 ± 0.5	1.043 (da)	0.734	39.58[h]	80.4	AIST (11/18)	Solar Frontier [9]
CIGSSe (submodule)	20.3 ± 0.4	526.7 (ap)	0.6834[c]	39.55[c,i]	75.1	NREL (5/23)	Avancis, 100 cells [10]
CdTe (cell)	21.0 ± 0.4	1.0623 (ap)	0.8759	30.25[g]	79.4	Newport (8/14)	First Solar, on glass [11]
CZTSSe (cell)	14.1 ± 0.3	1.075 (da)	0.5356	40.67[j]	69.7	NPVM (1/25)	IoP/CAS [12]
CZTSSe (minimodule)	**12.9 ± 0.3**	**10.51 (da)**	**0.5403**[c]	**33.84**[b,c]	**70.6**	**NPVM (7/25)**	**IoP/CAS, 6 cells** [12]
CZTS (cell)	10.0 ± 0.2	1.113 (da)	0.7083	21.77[g]	65.1	NREL (3/17)	UNSW [13]
Amorphous/microcrystalline							
Si (amorphous cell)	10.2 ± 0.3[f,k]	1.001 (da)	0.896	16.36[l]	69.8	AIST (7/14)	AIST [14]
Si (microcrystalline cell)	11.9 ± 0.3[f]	1.044 (da)	0.550	29.72[g]	75.0	AIST (2/17)	AIST [15]
Perovskite							
Perovskite (cell)	26.9 ± 0.8[m]	1.017 (da)	1.203	**26.67**[j]	**83.7**	NPVM (3/25)	Soochow/UNSW/BaimaLake [16]
Perovskite (minimodule)	23.9 ± 0.5[m]	19.48 (da)	1.070[c]	27.08[c,j]	82.6	NPVM (3/25)	Microquanta, 9 cells [17]
Perovskite (submodule)	**22.9 ± 0.5**[m]	**756.0 (da)**	**1.180**[c]	**24.31**[b,c]	**79.9**	**NPVM (9/25)**	**Mellow/JinanU, 45 cells** [18]
Dye sensitized							
Dye (cell)	11.9 ± 0.4[n]	1.005 (da)	0.744	22.47[o]	71.2	AIST (9/12)	Sharp [19, 20]
Dye (minimodule)	10.7 ± 0.4[n]	26.55 (da)	0.754[c]	20.19[c,p]	69.9	AIST (2/15)	Sharp, 7 serial cells [19, 20]
Dye (submodule)	8.8 ± 0.3[n]	398.8 (da)	0.697[c]	18.42[c,q]	68.7	AIST (9/12)	Sharp, 26 serial cells [19, 20]
Organic							

(Continues)

TABLE 1 | (Continued)

Classification	Efficiency (%)	Area (cm^2)	V_{oc} (V)	J_{sc} (mA/cm^2)	Fill factor (%)	Test center (date)	Description
Organic (cell)	**19.4 ± 0.4**[r]	**1.003 (da)**	**0.9341**	**26.31**[b]	**79.1**	**NPVM (9/25)**	**HyperPV** [21]
Organic (minimodule)	15.7 ± 0.3[r]	19.31 (da)	0.8771[c]	24.37[c,i]	73.4	JET (1/23)	ZhejiangU, 7 cells [22]
Organic (submodule)	14.5 ± 0.2[r]	204.11 (da)	0.8315[c]	23.32[c,s]	74.6	FhG-ISE (11/23)	FAU/FZJ, 38 cells [23]

Note: CIGS = $CuIn_{1-y}Ga_ySe_2$, CZTSSe = $Cu_2ZnSnS_{4-y}Se_y$, CZTS = Cu_2ZnSnS_4, (ap) = aperture area, (t) = total area, (da) = designated illumination area, ISFH = Institute für Solarenergieforschung, NREL = US National Renewable Energy Laboratory, FhG-ISE = Fraunhofer Institut für Solare Energiesysteme, AIST = Japanese National Institute of Advanced Industrial Science and Technology, NPVM = Chinese National Photovoltaic Industry Measurement and Testing Center, JET = Japan Electrical Safety and Environment Technology Laboratories.

[a]Contacting: front: unmetallized; Rear: 2x9BB, busbar resistance neglecting (brn) contacting, highly reflective chuck (hrc).
[b]Spectral response and current–voltage curve reported in the present version of these tables.
[c]Reported on a "per cell" basis.
[d]Recalibrated from original measurement.
[e]Spectral response and current–voltage curve reported in Version 53 of these tables.
[f]Not measured at an external laboratory.
[g]Spectral response and current–voltage curve reported in Version 50 of these tables.
[h]Spectral response and current–voltage curve reported in Version 54 of these tables.
[i]Spectral response and current–voltage curve reported in Version 62 of these tables.
[j]Spectral response and current–voltage curve reported in Version 66 of these tables.
[k]Stabilized by 1000 h exposure to 1 sun light at 50°C.
[l]Spectral response and current–voltage curve reported in Version 45 of these tables.
[m]Initial performance. References [24, 25] review the stability of similar devices.
[n]Initial efficiency. Reference [26] reviews the stability of similar devices.
[o]Spectral response and current–voltage curve reported in Version 41 of these tables.
[p]Spectral response and current–voltage curve reported in Version 46 of these tables.
[q]Spectral response and current–voltage curve reported in Version 43 of these tables.
[r]Initial performance. References [27, 28] review the stability of similar devices.
[s]Spectral response and current–voltage curve reported in Version 64 of these tables.

TABLE 2 | "Notable exceptions" for single-junction cells and submodules: "Top dozen" confirmed results, not class records, measured under the global AM1.5 spectrum (1000 W m^{-2}) at 25°C (IEC 60904-3: 2008 or ASTM G-173-03 global).

Classification	Efficiency (%)	Area (cm^2)	V_{oc} (V)	J_{sc} (mA/cm^2)	Fill factor (%)	Test center (date)	Description
Cells (silicon)							
Si (PERC)	25.0 ± 0.5	4.00 (da)	0.706	42.7[a]	82.8	Sandia (3/99)	UNSW, p-type [29]
Si (p-TOPCon)	26.0 ± 0.5[b]	4.015 (da)	0.7323	42.05[c]	84.3	FhG-ISE (11/19)	FhG-ISE, p-type [30]
Si (p-TBC)	26.1 ± 0.3[b]	3.9857 (da)	0.7266	42.62[d]	84.3	ISFH (2/18)	ISFH, p-type [31]
Si (large PERC)	24.1 ± 0.4[e]	441.3 (t)	0.6997	41.79[f]	82.3	ISFH (12/24)	Trina p-type [32]
Si (large TOPCon)	**26.55 ± 0.4[g]**	**335.2 (t)**	**0.7434**	**41.99[h]**	**85.1**	**ISFH (10/25)**	**Runergy, n-type** [33]
Si (large hybrid)	**26.6 ± 0.4[i]**	**165.81 (t)**	**0.7460**	**41.93[h]**	**85.1**	**ISFH (1/25)**	**LONGi, T-front, H-rear** [34]
Si (large HJT)	26.8 ± 0.4[j]	274.4 (t)	0.7514	41.45[k]	86.1	ISFH (10/22)	LONGi, n-type [35]
Si (large p-HJT)	26.6 ± 0.4[j]	274.1 (t)	0.7513	41.30[k]	85.6	ISFH (10/22)	LONGi, p-type [35]
Cells (chalcogenide)							
CIGS	23.6 ± 0.4	0.899 (da)	0.7671	38.30[l]	80.5	FhG-ISE (1/23)	Evolar/UppsalaU [36]
CdTe	23.1 ± 0.3	0.4507 (da)	0.9048	31.66[m]	80.6	NREL (5/24)	First Solar [37]
CZTSSe	**16.6 ± 0.4**	**0.228 (da)**	**0.5696**	**38.56[h]**	**75.4**	**NPVM (7/25)**	**IoP/CAS** [12]
Cells (other)							
Perovskite	27.3 ± 0.6[n]	0.1065 (da)	1.200	**26.64[f]**	**85.4**	NPVM (5/25)	Soochow/UNSW/ BaimaLake [16]
Dye sensitized	13.0 ± 0.4[o]	0.1155 (da)	1.0396	15.55[l]	80.4	FhG-ISE (10/20)	EPFL [38]

Note: CIGS = $CuIn_{1-y}Ga_ySe_2$, CZTSSe = $Cu_2ZnSnS_{4-y}Se_y$, CZTS = Cu_2ZnSnS_4, (ap) = aperture area, (t) = total area, (da) = designated illumination area, AIST, Japanese National Institute of Advanced Industrial Science and Technology, NREL, National Renewable Energy Laboratory, FhG-ISE, Fraunhofer-Institut für Solare Energiesysteme, ISFH, Institute for Solar Energy Research, Hamelin.
[a] Spectral response reported in Version 36 of these tables.
[b] Not measured at an external laboratory.
[c] Spectral response and current–voltage curves reported in Version 55 of these tables.
[d] Spectral response and current–voltage curve reported in Version 52 of these tables.
[e] Contacting: Front: 12BB, resistance neglecting (brn); Rear: fully metallized, full area contacting (fac).
[f] Spectral response and current–voltage curves reported in Version 66 of these tables.
[g] Contacting: 16BB, busbar resistance neglecting (brn); Rear: 10 continuous +10 dashed BBs, grid resistance neglecting (grn) contacting, highly reflective (gold) chuck (hrc).
[h] Spectral response and current–voltage curves reported in the present version of these tables.
[i] Contacting: Front: 18BB, busbar resistance neglecting (brn), only every 2nd BB contacted; Rear: fully metallized, full area contacting (fac).
[j] Contacting: Front: 12BB, busbar resistance neglecting (brn) contacting; Rear: 12BB, grid resistance neglecting (grn) contacting, highly reflective chuck (hrc).
[k] Spectral response and current–voltage curves reported in Version 61 of these tables.
[l] Spectral response and current–voltage curves reported in Version 62 of these tables.
[m] Spectral response and current–voltage curves reported Version 65 of these tables.
[n] Stability not investigated. References [24, 25] document stability of similar devices.
[o] Long term stability not investigated. Reference [26] documents stability of similar devices.

TABLE 3 | Confirmed nonconcentrating, terrestrial high-bandgap cell efficiencies measured under the global AM1.5 spectrum (1000 W m^{-2}) at 25°C (IEC 60904-3: 2008 or ASTM G-173-03 global).

Classification	E_G (eV)	Efficiency (%)	Area (cm^2)	V_{oc} (V)	J_{sc} (mA/cm^2)	Fill factor (%)	Test center (date)	Description
1.53–1.65 eV								
$Cu(Ga,In)S_2$	1.53	15.5 ± 0.3	0.4334 (ap)	0.920	23.41[a]	72.2	FhG-ISE (2015)	Showa Shell [39]
Perovskite[b]	1.57	24.2 ± 0.8	0.0955 (ap)	1.1948	24.16[c]	84.0	Newport (1/19)	KRICT [40]
CZTS	1.59	13.2 ± 0.3	0.2023 (da)	0.8342	22.16[d]	71.6	NPVM (5/24)	UNSW (Cd-free) [41]
$Cu(Ga,In)S_2$	**1.62**	**14.9 ± 0.9**	**0.198 (da)**	**0.8988**	**23.19[e]**	**71.5**	**NPVM (9/25)**	**UNSW (Cd-free)** [42]
1.65–1.75 eV								
Perovskite[b]	1.65	16.2 ± 0.6	<0.2 (da)	1.1085	19.65[c]	74.2	Newport (12/13)	KRICT [43]
$CuGaSe_2$:Al	1.67	12.25 ± 0.3[f]	0.4990 (da)	0.959	17.63[a]	72.5	AIST (1/24)	AIST [44]
1.75–1.85 eV								
GaInP	1.82	22.0 ± 0.3[f]	0.2502 (ap)	1.4695	16.63[g]	90.2	NREL (1/19)	NREL, rear HJ, strained AlInP [45]

Note: $Cu(Ga,In)S_2 = CuGa_{1-y}In_yS_2$, CZTS = Cu_2ZnSnS_4, (ap) = aperture area, (t) = total area, (da) = designated illumination area, AIST = Japanese National Institute of Advanced Industrial Science and Technology, NREL = National Renewable Energy Laboratory, FhG-ISE = Fraunhofer-Institut für Solare Energiesysteme, NPVM = Chinese National Photovoltaic Industry Measurement and Testing Center.
[a]Spectral response and current–voltage curves reported in Version 66 of these tables.
[b]Stability not investigated. References [24, 25] document stability of similar devices.
[c]Spectral response and more information reported in Ref. [42].
[d]Spectral response and current–voltage curves reported in Version 65 of these tables.
[e]Spectral response and current–voltage curves reported in the present version of these tables.
[f]Not measured at an external laboratory.
[g]Spectral response and current–voltage curve reported in Version 54 of these tables.

TABLE 4 | Confirmed multiple-junction terrestrial cell and submodule efficiencies measured under the global AM1.5 spectrum (1000 W/m^2) at 25°C (IEC 60904-3: 2008 or ASTM G-173-03 global).

Classification	Efficiency (%)	Area (cm^2)	V_{oc} (V)	J_{sc} (mA/cm^2)	Fill factor (%)	Test center (date)	Description
III–V multijunctions							
5 junction cell (bonded) (2.17/1.68/1.40/1.06/.73 eV)	38.8 ± 1.2	1.021 (ap)	4.767	9.564	85.2	NREL (7/13)	Spectrolab, 2-terminal
InGaP/GaAs/InGaAs	37.9 ± 1.2	1.047 (ap)	3.065	14.27[a]	86.7	AIST (2/13)	Sharp, 2 term [46].
GaInP/GaAs (monolithic)	32.8 ± 1.4	1.000 (ap)	2.568	14.56[b]	87.7	NREL (9/17)	LG Electronics, 2 term.
III–V/Si multijunctions							
GaInP/GaInAsP//Si (bonded)	36.1 ± 1.3[c]	3.987 (ap)	3.309	12.70[d]	86.0	FhG-ISE (5/23)	FhG-ISE/AMOLF, 2-term [47].
GaInP/GaAs/Si (mech. stack)	35.9 ± 0.5[c]	1.002 (da)	2.52/0.681	13.6/11.0	87.5/78.5	NREL (2/17)	NREL/CSEM/EPFL, 4-term [48].
GaInP/GaAs/Si (monolithic)	25.9 ± 0.9[c]	3.987 (ap)	2.647	12.21[e]	80.2	FhG-ISE (6/20)	Fraunhofer ISE, 2-term [49].
GaAsP/Si (monolithic)	23.4 ± 0.3	1.026 (ap)	1.732	17.34[f]	77.7	NREL (5/20)	OSU/UNSW/SolAero, 2-term [50]
GaAs/Si (mech. stack)	32.8 ± 0.5[c]	1.003 (da)	1.09/0.683	28.9/11.1[g]	85.0/79.2	NREL (12/16)	NREL/CSEM/EPFL, 4-term [48].
GaInP/GaInAs/Ge; Si (spectral split minimodule)	34.5 ± 2.0	27.83 (ap)	2.66/0.65	13.1/9.3	85.6/79.0	NREL (4/16)	UNSW/Azur/Trina, 4-term [51].
Perov./Si multijunctions							
Perovskite/Si	**35.0 ± 1.0**[h]	**1.0028 (da)**	**2.007**	**20.76**[i]	**84.3**	**ESTI (7/25)**	**LONGi, 2-term** [52].
Perovskite/Si (large)	**34.1 ± 1.2**[h]	**261.1 (ap)**	**2.013**	**19.51**[i]	**86.9**	**NREL (11/25)**	**LONGi, 2-term** [52].
Perovskite/Si (large)	28.6 ± 1.5[h]	330.56 (t)	1.903	18.94[j]	79.3	FhG-ISE (11/24)	HanwhaQCells, 2-term [53].
Other multijunctions							
Perovskite/CIGS	24.6 ± 1.1[h]	1.110 (da)	1.762	19.28[j]	72.5	FhG-ISE (12/24)	HZB, 2-terminal [54]
Perovskite/perovskite	28.2 ± 0.5[h]	1.038 (da)	2.159	16.59[k]	78.9	JET (12/22)	NanjingU/Renshine, 2-term [55].
Perovskite/perovskite (minimodule)	26.2 ± 0.6[h]	64.84 (da)	2.182	15.60[j]	77.4	JET (9/24)	Renshine/NanjingU, 14 cells [56]
a-Si/nc-Si/nc-Si (thin-film)	14.0 ± 0.4[c,l]	1.045 (da)	1.922	9.94[m]	73.4	AIST (5/16)	AIST, 2-term [57].
a-Si/nc-Si (thin-film)	12.7 ± 0.4[c,l]	1.000 (da)	1.342	13.45[n]	70.2	AIST (10/14)	AIST, 2-term [58].
"Notable exceptions"							
GaInP/GaAs (mqw)	32.9 ± 0.5[c]	0.250 (ap)	2.500	15.36[o]	85.7	NREL (1/20)	NREL/UNSW, multiple QW
GaInP/GaAs/GaInAs	37.8 ± 1.4	0.998 (ap)	3.013	14.60[o]	85.8	NREL (1/18)	Microlink (ELO) [59]
GaInP/GaAs (mqw)/GaInAs	39.5 ± 0.5[c]	0.242 (ap)	2.997	15.44[p]	85.3	NREL (9/21)	NREL, multiple QW
6 junction (monolithic) (2.19/1.76/1.45/1.19/0.97/.7 eV)	39.2 ± 3.2[c]	0.247 (ap)	5.549	8.457[q]	83.5	NREL (11/18)	NREL, inv. Metamorphic [60]

(Continues)

TABLE 4 | (Continued)

Classification	Efficiency (%)	Area (cm²)	V_{oc} (V)	J_{sc} (mA/cm²)	Fill factor (%)	Test center (date)	Description
GaInP/AlGaAs/CIGS	28.1 ± 1.2[c]	0.1386 (da)	2.952	11.72[r]	81.1	AIST (1/21)	AIST/FhG-ISE, 2-term [61].
Perovskite/perovskite	30.1 ± 0.8[h]	0.0493 (da)	2.20	16.72[s]	81.8	JET (10/23)	NanjingU/Renshine, 2-term [56].

Note: a-Si = amorphous silicon/hydrogen alloy, nc-Si = nanocrystalline or microcrystalline silicon, (ap) = aperture area, (t) = total area, (da) = designated illumination area, NREL = US National Renewable Energy Laboratory, AIST = Japanese National Institute of Advanced Industrial Science and Technology, FhG-ISE = Fraunhofer Institut für Solare Energiesysteme, ESTI = European Solar Test Installation, JET = Japan Electrical Safety and Environment Technology Laboratories.

[a]Spectral response and current–voltage curve reported in Version 42 of these tables.
[b]Spectral response and current–voltage curve reported in Version 51 of these tables.
[c]Not measured at an external laboratory.
[d]Spectral response and current–voltage curves reported in the present version of these tables.
[e]Spectral response and current–voltage curve reported in Version 57 of these tables.
[f]Spectral response and current–voltage curve reported in Version 56 of these tables.
[g]Spectral response and current–voltage curve reported in Version 52 of these tables.
[h]Initial efficiency. References [24, 25] review the stability of similar perovskite-based devices.
[i]Spectral response and current–voltage curves reported in the present version of these tables.
[j]Spectral response and current–voltage curve reported in Version 66 of these tables.
[k]Spectral response and current–voltage curve reported in Version 63 of these tables.
[l]Stabilized by 1000 h exposure to 1 sun light at 50°C.
[m]Spectral response and current–voltage curve reported in Version 49 of these tables.
[n]Spectral responses and current–voltage curve reported in Version 45 of these tables.
[o]Spectral response and current–voltage curve reported in Version 53 of these tables.
[p]Spectral response and current–voltage curves reported in Version 59 of these tables.
[q]Spectral response and current–voltage curve reported in Version 54 of these tables.
[r]Spectral response and current–voltage curve reported in Version 58 of these tables.
[s]Spectral response and current–voltage curve reported in Version 64 of these tables.

cells to encourage accelerated development of possible upper cell candidates in tandem cell stacks.

Highest confirmed "one sun" cell and module results are reported in Tables 1–5. Any changes in the tables from those previously published [1] are set in bold type. In most cases, a literature reference is provided that describes either the result reported or a similar result (readers identifying improved references are welcome to submit to the lead author). Table 1 summarizes the best-reported measurements for "one-sun" (nonconcentrator) single-junction cells and submodules.

Table 2 contains what might be described as "notable exceptions" for "one-sun" single-junction cells and submodules in the above category. While not conforming to the requirements to be recognized as a class record, the devices in Table 2 have notable characteristics that will be of interest to sections of the photovoltaic community, with entries based on their significance and timeliness. To encourage discrimination, the table is limited to nominally 12 entries, with the present authors having voted for their preferences for inclusion. Readers who have suggestions for notable exceptions for inclusion into this or subsequent tables are welcome to contact any of the authors with full details. Suggestions conforming to the guidelines will be included on the voting list for a future issue.

Table 3 is a new table introduced in Version 66 reporting highest confirmed results for cells of high bandgap (> 1.55 eV) and efficiency above 10% (see table heading and footnotes for other requirements). Table 4 (formerly Table 3) was first introduced in Version 49 of these tables and summarizes the growing number of cell and submodule results involving high efficiency, one-sun multiple-junction devices (even earlier, reported in Table 1). Table 5 (formerly Table 4) shows the best results for one-sun modules, both single- and multiple-junction, while Table 6 shows the best results for concentrator cells and concentrator modules. A small number of "notable exceptions" are also included in Tables 4–6.

2 | New Results

Seventeen new results and updates are reported in the present version of these tables. The first new entry reported in Table 1 ("one-sun cells and submodules") is a new efficiency record of 27.7% for a large total area (175-cm²) n-type silicon cell fabricated by LONGi Solar and measured by the Institute für Solarenergieforschung (ISFH) in October 2025. When the edge region was masked to minimize the impact of edge recombination, this efficiency increased slightly to 27.9%, the best outright result for a silicon cell. This cell uses an Interdigitated-Back-Contact (IBC) design, with TOPCon n-type and heterojunction p-type contacts, making it a HTBC cell in the common designation.

Another new result in Table 1 is an energy conversion efficiency of 12.9% reported for a 10.5-cm² $Cu_2ZnSnS_ySe_{4-y}$ (CZTSSe) minimodule fabricated by the Institute of Physics, Chinese Academy of Sciences (IoP/CAS) [12] and measured by the Chinese National Photovoltaic Industry Measurement and Testing Center (NPVM), improving on the Institute's earlier 11.95%

TABLE 5 | Confirmed nonconcentrating terrestrial module efficiencies measured under the global AM1.5 spectrum (1000 W/m^2) at a cell temperature of 25°C (IEC 60904-3: 2008 or ASTM G-173-03 global).

Classification	Effic. (%)	Area (cm^2)	V_{oc} (V)	I_{sc} (A)	FF (%)	Test center (date)	Description
Si (crystalline)	26.0 ± 0.3	18,156 (da)	40.38	13.896[a]	84.0	NREL (12/24)	LONGI, HBC [5]
Si (crystalline)	25.4 ± 0.4	16,279 (ap)	56.09	8.58[a]	86.0	FhG-ISE (12/24)	Trina, HJT [62]
GaAs (thin-film)	25.1 ± 0.8	866.45 (ap)	11.08	2.303[b]	85.3	FhG-ISE (11/17)	Alta Devices [63]
CIGS (Cd-free)	19.2 ± 0.5	841 (ap)	48.0	0.456[c]	73.7	AIST (1/17)	Solar Frontier (70 cells) [64]
CdTe (thin-film)	19.9 ± 0.3	23,932 (da)	231.5	2.675[d]	77.1	NREL (6/23)	First Solar [65]
Perovskite	**21.1 ± 0.4[e]**	**843.5 (da)**	**56.5**	**0.420[f]**	**74.9**	**NREL (11/24)**	**SolaEon** [66]
Organic	13.1 ± 0.3[g]	1475 (da)	48.10	0.6015[h]	67.0	NREL (5/23)	Waystech/Nanobit [67]
Multijunction							
InGaP/GaAs/InGaAs	32.65 ± 0.7	965 (da)	24.30	1.520[i]	85.3	AIST (2/22)	Sharp (40 cells; 8 series) [68]
Perovskite/Si	**31.1 ± 0.9[e]**	**1600.6 (da)**	**11.94**	**5.056[f]**	**82.5**	**FhG-ISE (7/25)**	**LONGi** [52]
Perovskite/Si (large)	**29.1 ± 0.8[e]**	**16,820 (da)**	**59.54**	**10.271[f]**	**80.0**	**FhG-ISE (7/25)**	**LONGi** [52]
Perovskite/perovskite	**22.8 ± 0.8[e]**	**809.7 (da)**	**90.24**	**0.282[a]**	**72.6.1**	**FhG-ISE (4/25)**	**RenShine Solar** [69]
a-Si/nc-Si (tandem)	12.3 ± 0.3[j]	14,322 (t)	280.1	0.902[k]	69.9	ESTI (9/14)	TEL Solar, Trubbach Labs [70]
"Notable exceptions"							
CIGS (large)	18.6 ± 0.6	10,858 (ap)	58.00	4.545[l]	76.8	FhG-ISE (10/19)	Miasole [71]
InGaP/GaAs//Si	33.7 ± 0.7	775 (da)	20.3/2.83	1.25/1.93[h]	86.5/78.0	AIST (2/23)	Sharp/Toyota TI, 4-term [72].
InGaP/GaAs//CIGS	31.2 ± 0.7	778 (ap)	20.3/16.9	1.24/.26[h]	85.7/59.8	AIST (2/23)	Sharp/Idemitsu, 4-term [72].
Perovskite (large)	18.1 ± 0.6[e]	7218 (t)	93.56	1.876[a]	74.4	NREL (1/25)	UtmoLight [73]

Note: CIGSS = CuInGaSSe, a-Si = amorphous silicon/hydrogen alloy, a-SiGe = amorphous silicon/germanium/hydrogen alloy, nc-Si = nanocrystalline or microcrystalline silicon, Effic. = efficiency, (t) = total area, (ap) = aperture area, (da) = designated illumination area, FF = fill factor.

[a]Spectral response and current–voltage curve reported in Version 66 of these tables.
[b]Spectral response and current–voltage curve reported in Version 51 of these tables.
[c]Spectral response and current–voltage curve reported in Version 50 of these tables.
[d]Spectral response and current–voltage curve reported in Version 65 of these tables.
[e]Initial performance. References [24, 25] review the stability of similar devices.
[f]Spectral response and current–voltage curve reported in the present version of these tables.
[g]Initial performance. References [27, 28] review the stability of similar devices.
[h]Spectral response and current voltage curve reported Version 62 of these tables.
[i]Spectral response and current–voltage curve reported in Version 60 of these tables.
[j]Stabilized at the manufacturer to the 2% level following IEC procedure of repeated measurements.
[k]Spectral response and/or current–voltage curve reported in Version 46 of these tables.
[l]Spectral response and current–voltage curve reported in Version 55 of these tables.

TABLE 6 | Terrestrial concentrator cell and module efficiencies measured under the ASTM G-173-03 direct beam AM1.5 spectrum at a cell temperature of 25°C (except where noted for the hybrid and luminescent modules).

Classification	Effic. (%)	Area (cm^2)	Intensity[a] (suns)	Test center (date)	Description
Single cells					
GaAs	30.8 ± 1.9[b,c]	0.0990 (da)	61	NREL (1/22)	NREL, 1 junction (1J)
Si	27.6 ± 1.2[d]	1.00 (da)	92	FhG-ISE (11/04)	Amonix back-contact [74]
CIGS (thin-film)	23.3 ± 1.2[b,e]	0.09902 (ap)	15	NREL (3/14)	NREL [75]
Multijunction cells					
AlGaInP/AlGaAs/GaAs/GaInAs(3) (2.15/1.72/1.41/1.17/0.96/0.70 eV)	47.1 ± 2.6[b,f]	0.099 (da)	143	NREL (3/19)	NREL, 6J inv. Metamorphic [60]
GaInP/GaInAs; GaInAsP/GaInAs	47.6 ± 2.6[b,g]	0.0452 (da)	665	FhG-ISE (5/22)	FhG-ISE 4J bonded [76]
GaInP/GaAs/GaInAs/GaInAs	45.7 ± 2.3[b,h]	0.09709 (da)	234	NREL (9/14)	NREL, 4J monolithic [77]
InGaP/GaAs/InGaAs	44.4 ± 2.6[i]	0.1652 (da)	302	FhG-ISE (4/13)	Sharp, 3J inverted metamorphic [78]
GaInAsP/GaInAs	35.5 ± 1.2[b,j]	0.10031 (da)	38	NREL (10/17)	NREL 2-junction (2J) [79]
Minimodule					
GaInP/GaAs; GaInAsP/GaInAs	43.4 ± 2.4[b,k]	18.2 (ap)	340[l]	FhG-ISE (7/15)	Fraunhofer ISE 4J (lens/cell) [80]
Submodule					
GaInP/GaInAs/Ge; Si	40.6 ± 2.0[k]	287 (ap)	365	NREL (4/16)	UNSW 4J split spectrum [81]
Modules					
Si	20.5 ± 0.8[b]	1875 (ap)	79	Sandia (4/89)[l]	Sandia/UNSW/ENTECH (12 cells) [82]
Three junction (3J)	35.9 ± 1.8[m]	1092 (ap)	N/A	NREL (8/13)	Amonix [83]
Four junction (4J)	38.9 ± 2.5[n]	812.3 (ap)	333	FhG-ISE (4/15)	Soitec [84]
Hybrid module[o]					
4-junction (4J)/bifacial c-Si	34.2 ± 1.9[b,o]	1088 (ap)	CPV/PV	FhG-ISE (9/19)	FhG-ISE (48/8 cells; 4 T) [85]
"Notable exceptions"					

(Continues)

TABLE 6 | (Continued)

Classification	Effic. (%)	Area (cm^2)	Intensity[a] (suns)	Test center (date)	Description
Si (large area)	21.7 ± 0.7	20.0 (da)	11	Sandia (9/90)[l]	UNSW laser grooved [86]
Luminescent minimodule[o]	7.1 ± 0.2	25 (ap)	2.5[p]	ESTI (9/08)	ECN Petten, GaAs cells [87]
4J minimodule	41.4 ± 2.6[b]	121.8 (ap)	230	FhG-ISE (9/18)	FhG-ISE, 10 cells [88]

Note: Following the normal convention, efficiencies calculated under this direct beam spectrum neglect the diffuse sunlight component that would accompany this direct spectrum. These direct beam efficiencies need to be multiplied by a factor estimated as 0.8746 to convert to thermodynamic efficiencies [89].

CIGS = $CuInGaSe_2$, Effic. = efficiency, (da) = designated illumination area, (ap) = aperture area, NREL = National Renewable Energy Laboratory, FhG-ISE = Fraunhofer-Institut für Solare Energiesysteme, ESTI = European Solar Test Installation.

[a] One sun corresponds to direct irradiance of 1000 $W\,m^{-2}$.
[b] Not measured at an external laboratory.
[c] Spectral response and current–voltage curve reported in Version 60 of these tables.
[d] Measured under a low aerosol optical depth spectrum similar to ASTM G-173-03 direct [90].
[e] Spectral response and current–voltage curve reported in Version 44 of these tables.
[f] Spectral response and current–voltage curve reported in Version 54 of these tables.
[g] Spectral response and current–voltage curve reported in Version 61 of these tables.
[h] Spectral response and current–voltage curve reported in Version 46 of these tables.
[i] Spectral response and current–voltage curve reported in Version 42 of these tables.
[j] Spectral response and current–voltage curve reported in Version 51 of these tables.
[k] Determined at IEC 62670-1 CSTC reference conditions.
[l] Recalibrated from original measurement.
[m] Referenced to 1000 W/m^2 direct irradiance and 25°C cell temperature using the prevailing solar spectrum and an in-house procedure for temperature translation.
[n] Measured under IEC 62670-1 reference conditions following the current IEC power rating draft 62670-3.
[o] Thermodynamic efficiency. Hybrid and luminescent modules measured under the ASTM G-173-03 or IEC 60904-3: 2008 global AM1.5 spectrum at a cell temperature of 25°C. Four-terminal module with external dual-axis tracking. Power rating of CPV follows IEC 62670-3 standard, front power rating of flat plate PV based on IEC 60904-3, −5, −7, −10, and 60891 with modified current translation approach; rear power rating of flat plate PV based on IEC TS 60904-1-2 and 60891.
[p] Geometric concentration.

result [1]. A further new result is a large increase in efficiency to 22.9% reported for a 756-cm^2 Pb-halide perovskite submodule involving 45 cells in series fabricated by Mellow Energy in conjunction with Jinan University and again measured by NPVM. This submodule area is only slightly smaller than required for classification as a module.

The final new result in Table 1 is a large jump in efficiency to 19.4% for a 1-cm^2 organic solar cell fabricated by Hyper PV Technology Company Limited of Jiaxing, China and once more measured by NPVM. A recent compilation of organic cell efficiencies suggests that no recent publications claim efficiency over 19% for any cell of area larger than 0.1 cm^2, highlighting the significance of this result.

Three new results and one update are reported in Table 2 (one-sun "notable exceptions"). The first two are in the large area Si cell category where a change was introduced in the previous issue of the tables [1]. To allow a more direct comparison between the potential of the commercial technologies, they need to be reported on the same area basis, with "total area" assessed as the most sensible. The first of the new entries is 26.5% efficiency for a 335-cm^2 TOPCon cell fabricated by Jinko Solar and measured by ISFH. The second new entry is 26.6% for a 166-cm^2 n-type, rear junction hybrid cell with front TOPCon n-type contacts and rear HJT p-type contacts fabricated by LONGi Solar and again measured by ISFH.

The final new Table 2 entry is 16.6% for a 0.2-cm^2 CZTSSe cell fabricated by IoP/CAS [11] and measured by NPVM, an improvement over the institute's earlier 15.8% result. Finally, a correction is made to the current density and fill factor of the 27.3% Pb-halide perovskite solar cell (0.1 cm^2) fabricated by Soochow University in conjunction with UNSW Sydney, also measured by NPVM. For these last two results, cell area is too small for classification as an outright record, with solar cell efficiency targets in governmental research programs generally specified in terms of a cell area of 1 cm^2 or larger [91–93].

Table 3 was introduced in the previous issue of the tables and shows entries for high "effective bandgap" (E_G) cells (nominally > 1.55 eV). Such effective bandgaps are potential candidates for upper cells in tandem cell stacks. Because cell external quantum efficiency (EQE) is available for every certified cell measurement, a procedure with sound underpinnings [94] as described below is used to extract E_G from EQE. The sole new entry is 14.9% efficiency for a small area 0.2-cm^2 $Cu(Ga,In)S_2$ cell with 1.62-eV E_G fabricated by UNSW Sydney and measured at NPVM.

There are two new results reported in Table 4 describing results for one-sun, multijunction devices—both involving perovskite/

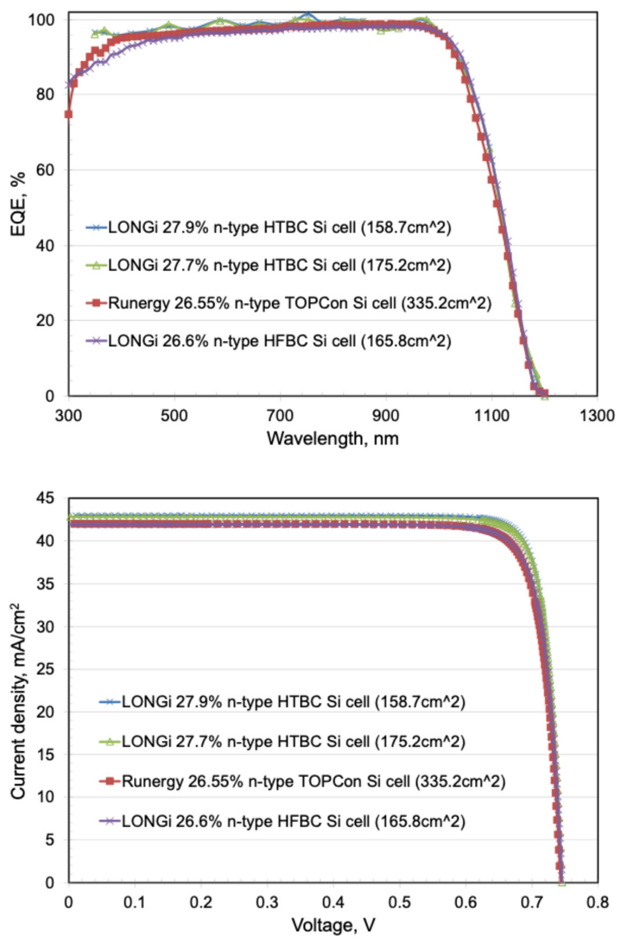

FIGURE 1 | (a) External quantum efficiency (EQE) for the new silicon cell results reported in this issue (some curves are normalized). (b) Corresponding current density–voltage (JV) curves.

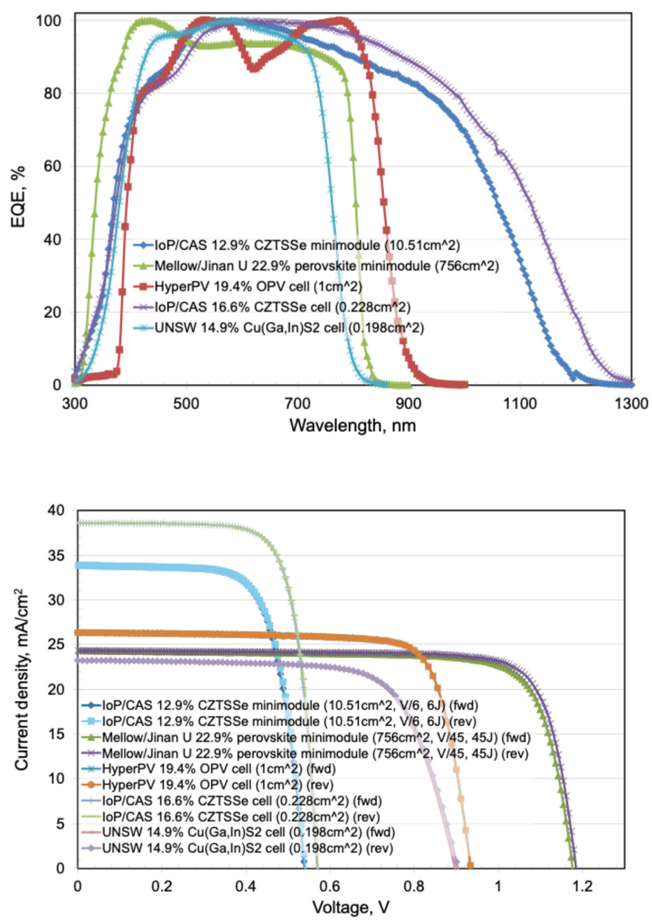

FIGURE 2 | (a) External quantum efficiency (EQE) for new thin-film cell and minimodule results reported in this issue (curves are normalized). (b) Corresponding current density–voltage (JV) curves.

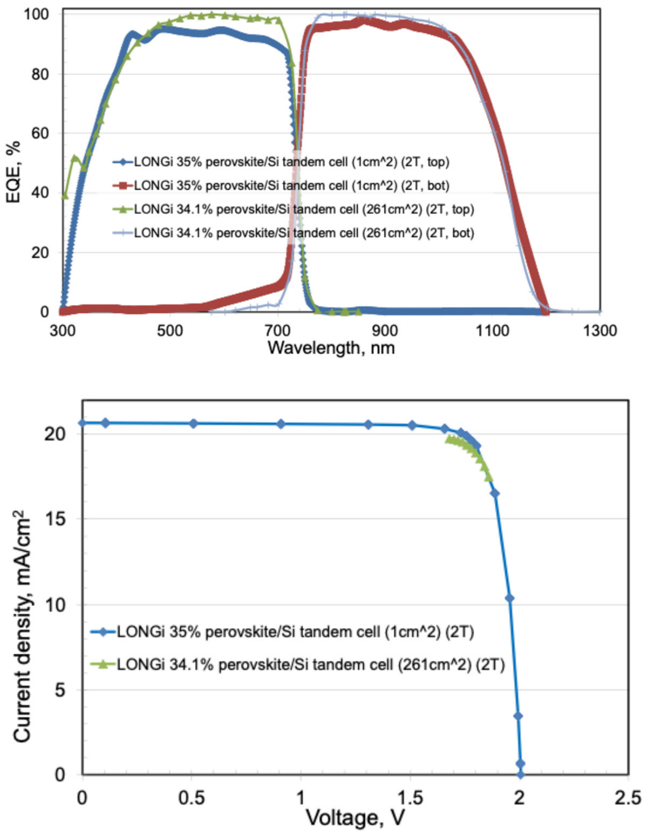

FIGURE 3 | (a) External quantum efficiency (EQE) for the new perovskite/silicon tandem cell results reported in this issue (some curves are normalized). (b) Corresponding current density–voltage (JV) curves.

silicon tandem cell stacks. An efficiency of 35.0% is reported for a 1-cm^2, two-terminal, silicon/perovskite tandem cell fabricated by LONGi Central R&D Institute and confirmed by the European Solar Test Installation (ESTI), beating out LONGi's earlier 34.85% result [1]. The second is an aperture area efficiency of 34.1% for a much larger area (261-cm^2), two-terminal, silicon/perovskite tandem cell again fabricated by LONGi [52] and measured by the US National Renewable Energy Laboratory (NREL).

There are four new results reported in Table 5 (one-sun modules) all involving perovskite technologies. The first is an efficiency of 21.1% reported for a small 843-cm^2 perovskite module fabricated by SolaEon and measured by NREL, displacing SolaEon's earlier 19.2% result. The second result is improvement to 31.1% efficiency for a small (0.16-m^2) perovskite/silicon module involving six tandem cells fabricated by LONGi and measured by the Fraunhofer Institute for Solar Energy Conversion (FhG-ISE). The third result is improvement to 29.1% efficiency for a much larger 1.7-m^2 perovskite/silicon module again fabricated by LONGi and measured by FhG-ISE.

The final new result is 22.8% for a small 810-cm^2 perovskite/perovskite tandem module fabricated by RenShine Solar and measured by FhG-ISE. There are no new entries in Table 6.

The EQE spectra for the new silicon cells reported in the present issue of these Tables are shown in Figure 1a, with Figure 1b

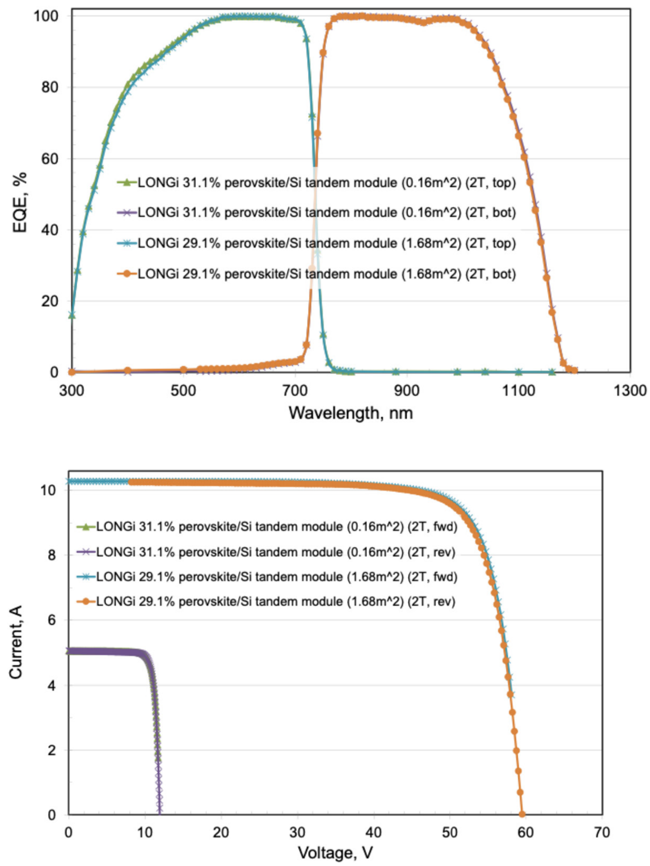

FIGURE 4 | (a) External quantum efficiency (EQE) for new two-terminal, double-junction, perovskite/silicon module results reported in this issue (results are normalized). (b) Corresponding current–voltage (IV) curves (different colors for each device represent forward and reverse characteristics).

showing the current density–voltage (JV) curves for the same devices. Figure 2a,b shows the corresponding EQE and JV curves for several of the new thin-film cell and minimodule results. Figure 3a,b shows these for the new two-terminal double-junction, perovskite/silicon tandem cell results and minimodule results, while Figure 4a,b shows these for the new perovskite/silicon tandem module results.

Author Contributions

All authors provided data and/or reviewed these and the manuscript. M.A.G. and J.Y.J. compiled the data and wrote the manuscript.

Acknowledgments

The Australian Centre for Advanced Photovoltaics commenced operation in February 2013 with support from the Australian Government through the Australian Renewable Energy Agency (ARENA). The Australian Government does not accept responsibility for the views, information or advice expressed herein. The work at NREL was supported by the US Department of Energy under Contract No. DE-AC36-08-GO28308 with the National Renewable Energy Laboratory. The work at AIST was supported in part by the Japanese New Energy and Industrial Technology Development Organisation (NEDO) under the Ministry of Economy, Trade and Industry (METI).

Funding

This study was supported by the Australian Renewable Energy Agency, Grant/Award Number: SRI-001; the U.S. Department of Energy (Office of Science, Office of Basic Energy Sciences and Energy Efficiency and Renewable Energy, Solar Energy Technology Program), Grant/Award Number: DE-AC36-08-GO28308; and the Ministry of Economy, Trade and Industry.

Disclosure

While the information provided in the tables is provided in good faith, the authors, editors, and publishers cannot accept direct responsibility for any errors or omissions.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author.

References

1. M. A. Green, E. D. Dunlop, M. Yoshita, et al., "Solar Cell Efficiency Tables (Version 66)," *Progress in Photovoltaics: Research and Applications* 33 (2025): 795–810.

2. M. A. Green, E. D. Dunlop, G. Siefer, et al., "Solar Cell Efficiency Tables (Version 61)," *Progress in Photovoltaics: Research and Applications* 31 (2022): 3–16.

3. M. A. Green, E. D. Dunlop, J. Hohl-Ebinger, et al., "Solar Cell Efficiency Tables (Version 60)," *Progress in Photovoltaics: Research and Applications* 30 (2022): 687–701.

4. M. A. Green, K. Emery, Y. Hishikawa, and W. Warta, "Solar Cell Efficiency Tables (Version 33)," *Progress in Photovoltaics: Research and Applications* 17 (2009): 85–94.

5. G. Wang, M. Yu, H. Wu, et al., "Silicon Solar Cells With Hybrid Back Contacts," *Nature* 649 (2025): 369–374.

6. M. J. Keevers, T. L. Young, U. Schubert, and M. A. Green. "10% Efficient CSG Minimodules," 22nd European Photovoltaic Solar Energy Conference, Milan, September 2007.

7. B. M. Kayes, H. Nie, R. Twist, et al., "27.6% Conversion Efficiency, a New Record for Single-Junction Solar Cells Under 1 Sun Illumination," in *Proceedings of the 37th IEEE Photovoltaic Specialists Conference*, 2011.

8. M. Wanlass. "Systems and Methods for Advanced Ultra-High-Performance," in P solar cells. US Patent 9,590,131 B2 2017.

9. M. Nakamura, K. Yamaguchi, Y. Kimoto, Y. Yasaki, T. Kato, and H. Sugimoto, "Cd-Free Cu(In,Ga)(Se,S)$_2$ Thin-Film Solar Cell With a New World Record Efficacy of 23.35%," 2019 46th IEEE PVSC, Chicago, IL (see also http://www.solar-frontier.com/eng/news/2019/0117_press.html).

10. R. Diermann, "Avancis Claims 19.64% Efficiency for CIGS Module, PV Magazine International," 2021, https://www.pv-magazine.com/2021/03/04/avancis-claims-19-64-efficiency-for-cigs-module/.

11. First Solar Press Release, "First Solar Builds the Highest Efficiency Thin Film PV Cell on Record," 2014.

12. J. Zhou, X. Xu, H. Wu, et al., "Control of the Phase Evolution of Kesterite by Tuning of the Selenium Partial Pressure for Solar Cells With 13.8% Certified Efficiency," *Nature Energy* 8 (2023): 526–535.

13. C. Yan, J. Huang, K. Sun, et al., "$Cu_2ZnSn\ S_4$ Solar Cells With Over 10% Power Conversion Efficiency Enabled by Heterojunction Heat Treatment," *Nature Energy* 3 (2018): 764–772.

14. T. Matsui, A. Bidiville, H. Sai, et al., "High-Efficiency Amorphous Silicon Solar Cells: Impact of Deposition Rate on Metastability," *Applied Physics Letters* 106 (2015): 053901, https://doi.org/10.1063/1.4907001.

15. H. Sai, T. Matsui, H. Kumagai, and K. Matsubara, "Thin-Film Microcrystalline Silicon Solar Cells: 11.9% Efficiency and Beyond," *Applied Physics Express* 11 (2018): 022301.

16. https://www.acap.org.au/post/world-leading-27-perovskite-efficiency-record-achieved-by-unsw-and-soochow-university-with-acap-su.

17. https://www.perovskite-info.com/microquanta-announces-2412-efficiency-1948cm-perovskite-solar-modules.

18. https://www.solarbeglobal.com/mellow-energy-achieves-record-22-86-efficiency-in-large-area-perovskite-solar-modules/.

19. L. Han, A. Fukui, Y. Chiba, et al., "Integrated Dye-Sensitized Solar Cell Module With Conversion Efficiency of 8.2%," *Applied Physics Letters* 94 (2009): 013305, https://doi.org/10.1063/1.3054160.

20. R. Komiya, A. Fukui, N. Murofushi, N. Koide, R. Yamanaka, and H. Katayama, "Improvement of the Conversion Efficiency of a Monolithic Type Dye-Sensitized Solar Cell Module," 2011 Technical Digest, 21st International Photovoltaic Science and Engineering Conference, Fukuoka; 2C-5O-08.

21. http://www.hyperchemical.com/services.html.

22. https://www.pv-magazine.com/2021/03/17/organic-pv-module-with-12-36-efficiency/.

23. R. Basu, F. Gumpert, J. Lohbreier, et al., "Large-Area Organic Photovoltaic Modules With 14.5% Certified World Record Efficiency," *Joule* 8 (2024): 970–978.

24. C. C. Boyd, R. Cheacharoen, T. Leijtens, and M. D. McGehee, "Understanding Degradation Mechanisms and Improving Stability of Perovskite Photovoltaics," *Chemical Reviews* 119 (2019): 3418–3451.

25. Y. Yang and J. You, "Make Perovskite Solar Cells Stable," *Nature* 544 (2017): 155–156.

26. U. O. Krašovec, M. Bokalič, M. Topič, "Ageing of DSSC Studied by Electroluminescence and Transmission Imaging," *Solar Energy Materials and Solar Cells* 117 (2013): 67–72.

27. D. M. Tanenbaum, M. Hermenau, E. Voroshazi, et al., "The ISOS-3 Inter-Laboratory Collaboration Focused on the Stability of a Variety of Organic Photovoltaic Devices," *RSC Advances* 2 (2012): 882–893.

28. M. Jørgensen, K. Norrman, S. A. Gevorgyan, T. Tromholt, B. Andreasen, and F. C. Krebs, "Stability of Polymer Solar Cells," *Advanced Materials* 24 (2012): 580–612.

29. M. A. Green, "The Passivated Emitter and Rear Cell (PERC): From Conception to Mass Production," *Solar Energy Materials and Solar Cells* 143 (2015): 190–197.

30. A. Armin Richter, R. Mueller, J. Benick, et al., "Design Rules for High-Efficiency Both-Sides Contacted Silicon Solar Cells With Balanced Charge Carrier Transport and Recombination Losses," *Nature Energy* 6 (2021): 429–438.

31. F. Haase, C. Klamt, S. Schäfer, et al., "Laser Contact Openings for Local Poly-Si-Metal Contacts Enabling 26.1%-Efficient POLO-IBC Solar Cells," *Solar Energy Materials and Solar Cells* 186 (2018): 184–193.

32. https://www.pv-magazine.com/2022/07/13/trina-solar-achieves-24-5-efficiency-for-210-mm-p-type-perc-solar-cell/.

33. https://www.runergy.com/runergy-achieves-26-55-full-area-n-type-cell-efficiency-setting-a-new-world-record/.

34. H. Lin, M. Yang, X. Ru, et al., "Silicon Heterojunction Solar Cells With Up to 26.81% Efficiency Achieved by Electrically Optimized Nanocrystalline-Silicon Hole Contact Layers," *Nature Energy* 8 (2023): 789–799.

35. "LONGi Achieves New World Record for p-Type Solar Cell Efficiency," Press Release. 2022, https://www.longi.com/en/news/p-type-hjt-record/.

36. K. Keller, K. Kiselman, O. Donzel-Gargand, et al., "High-Concentration Silver Alloying and Steep Back-Contact Gallium Grading Enabling Copper Indium Gallium Selenide Solar Cell With 23.6% Efficiency," *Nature Energy* 9 (2024): 467–478.

37. First Solar Press Release, First Solar Achieves yet Another Cell Conversion Efficiency World Record, 2016.

38. Y. Ren, D. Zhang, J. Suo, et al., "Hydroxamic Acid Pre-Adsorption Raises the Efficiency of Cosensitized Solar Cells," *Nature* 613, no. 7942 (2023): 60–65.

39. H. Hiroi, Y. Iwata, S. Adachi, H. Sugimoto, and A. Yamada, "New World-Record Efficiency for Pure-Sulfide Cu(In,Ga)S_2 Thin-Film Solar Cell With Cd-Free Buffer Layer via KCN-Free Process," *IEEE Journal of Photovoltaics* 6, no. 3 (2016): 760–763.

40. E. H. Jung, N. J. Jeon, E. Y. Park, et al., "Efficient, Stable and Scalable Perovskite Solar Cells Using Poly(3-Hexylthiophene)," *Nature* 567, no. 7749 (2019): 511–515.

41. A. Wang, J. Cong, S. Zhou, et al., "Hydrogen-Enhanced Carrier Collection Enabling Wide-Bandgap Cd-Free Cu_2ZnSnS_4 Solar Cells With 11.4% Certified Efficiency," *Nature Energy* 10 (2025): 255–265.

42. G. He, C. Yan, J. Li, et al., "11.6% Efficient Pure Sulfide Cu(In,Ga)S_2 Solar Cell Through a Cu-Deficient and KCN-Free Process," *ACS Applied Energy Materials* 3 (2020): 11974–11980.

43. M. A. Green and A. W. Y. Ho-Baillie, "Pushing to the Limit: Radiative Efficiencies of Recent Mainstream and Emerging Solar Cells," *ACS Energy Letters* 4 (2019): 1639–1644.

44. S. Ishizuka, J. Nishinaga, Y. Kamikawa, T. Nishida, and P. J. Fons, "Photovoltaic Efficiency Enhancement of Indium-Free Wide-Bandgap Chalcopyrite Solar Cells via an Aluminum-Induced Back-Surface Field Effect," *ACS Applied Materials & Interfaces* 17 (2025): 3136–3145.

45. NREL, private communication, 22 May 2019.

46. K. Sasaki, T. Agui, K. Nakaido, N. Takahashi, R. Onitsuka, and T. Takamoto, "Proceedings, 9th International Conference on Concentrating Photovoltaics Systems," Miyazaki, Japan 2013.

47. P. Schygulla, R. Müller, O. Höhn, et al., "Wafer-Bonded Two-Terminal III-V//Si Triple-Junction Solar Cell With Power Conversion Efficiency of 36.1% at AM1.5g," Presented at the 40th EU PVSEC, Lisbon, Paper 2DO.9 (submitted to Prog.Photovolt. Res. Appl.), 2023.

48. S. Essig, C. Allebé, T. Remo, et al., "Raising the One-Sun Conversion Efficiency of III–V/Si Solar Cells to 32.8% for Two Junctions and 35.9% for Three Junctions," *Nature Energy* 2 (2017): 17144, https://doi.org/10.1038/nenergy.2017.144.

49. M. Feifel, D. Lackner, J. Schön, et al., "Epitaxial GaInP/GaAs/Si Triple-Junction Solar Cell With 25.9% AM1.5g Efficiency Enabled by Transparent Metamorphic $Al_xGa_{1-x}As_yP_{1-y}$ Step-Graded Buffer Structures," *Solar RRL* 5 (2021): 2000763, https://doi.org/10.1002/solr.202000763.

50. T. J. Grassman, D. J. Chmielewski, S. D. Carnevale, J. A. Carlin, and S. A. Ringel, "$GaAs_{0.75}P_{0.25}$/Si Dual-Junction Solar Cells Grown by MBE and MOCVD," *IEEE Journal of Photovoltaics* 6 (2016): 326–331.

51. M. A. Green, M. J. Keevers, B. Concha Ramon, et al., "Improvements in Sunlight to Electricity Conversion Efficiency: Above 40% for Direct Sunlight and Over 30% for Global," Paper 1AP.1.2, European Photovoltaic Solar Energy Conference 2015, Hamburg, 2015.

52. https://www.longi.com/en/feature-report/world-record-for-solar-cell-efficiency/.

53. https://www.solarpowerworldonline.com/2024/12/qcells-reaches-28-6-efficiency-on-full-size-tandem-perovskite-silicon-solar-cell/.

54. G. Farias-Basulto, T. Mehlhop, N. J. Otto, et al., "Improving Perovskite/CIGS Tandem Solar Cells for Higher Power Conversion Efficiency Through Light Management and Bandgap Engineering," *ACS Applied Materials & Interfaces* 17 (2025): 56250–56255, https://doi.org/10.1021/acsami.5c15458.

55. R. Lin, J. Xu, M. Y. Wei, et al., "All-Perovskite Tandem Solar Cells With Improved Grain Surface Passivation," *Nature* 603 (2022): 73–78.

56. H. Gao, K. Xiao, R. Lin, et al., "Homogeneous Crystallization and Buried Interface Passivation for Perovskite Tandem Solar Modules," *Science* 383 (2024): 855–859, https://doi.org/10.1126/science.adj6088.

57. H. Sai, T. Matsui, T. Koida, and K. Matsubara, "Stabilized 14.0%-Efficient Triple-Junction Thin-Film Silicon Solar Cell," *Applied Physics Letters* 109 (2016): 183506, https://doi.org/10.1063/1.49669996.

58. T. Matsui, K. Maejima, A. Bidiville, et al., "High-Efficiency Thin-Film Silicon Solar Cells Realized by Integrating Stable a-Si:H Absorbers Into Improved Device Design," *Japanese Journal of Applied Physics* 54 (2015): 08KB10, https://doi.org/10.7567/JJAP.54.08KB10.

59. http://mldevices.com/index.php/news/ accessed October 28, 2018.

60. J. F. Geisz, M. A. Steiner, N. Jain, et al., "Building a Six-Junction Inverted Metamorphic Concentrator Solar Cell," *IEEE Journal of Photovoltaics* 8 (2018): 626–632.

61. K. Makita, Y. Kamikawa, H. Mizuno, et al., "III-V//$Cu_xIn_{1-y}Ga_ySe_2$ Multijunction Solar Cells With 27.2% Efficiency Fabricated Using Modified Smart Stack Technology With Pd Nanoparticle Array and Adhesive Material," *Progress in Photovoltaics: Research and Applications* 29 (2021): 887–898, https://doi.org/10.1002/pip.3398.

62. https://www.pv-tech.org/trina-solar-sets-25-44-fully-passivated-hjt-solar-module-efficiency-record/.

63. L. S. Mattos, S. R. Scully, M. Syfu, et al., "New Module Efficiency Record: 23.5% Under 1-Sun Illumination Using Thin-Film Single-Junction GaAs Solar Cells," *Proceedings of the 38th IEEE Photovoltaic Specialists Conference* (2012).

64. H. Sugimoto, "High Efficiency and Large Volume Production of CIS-Based Modules," 40th IEEE Photovoltaic Specialists Conference, Denver (2014).

65. http://www.firstsolar.com/en-AU/-/media/First-Solar/Technical-Documents/Series-6-Datasheets/Series-6-Datasheet.ashx, accessed October 28, 2019.

66. https://taiyangnews.info/solaeon-technology-announces-world-record-for-perovskite-modules/.

67. M Hosoya, H Oooka, H Nakao, et al., "Organic Thin Film Photovoltaic Modules," *Proceedings of the 93rd Annual Meeting of the Chemical Society of Japan* (2013): 21–37.

68. "Sharp Achieves World's Highest Conversion Efficiency of 32.65% in a Lightweight, Flexible, Practically Sized Solar Module," Press Release: June 6, 2022, https://global.sharp/corporate/news/220606-a.html.

69. https://www.oxfordpv.com/news/oxford-pv-debuts-residential-solar-module-record-setting-269-efficiency.

70. J. S. Cashmore, M. Apolloni, A. Braga, et al., "Improved Conversion Efficiencies of Thin-Film Silicon Tandem (MICROMORPH) Photovoltaic Modules," *Solar Energy Materials and Solar Cells* 144 (2016): 84–95, https://doi.org/10.1016/j.solmat.2015.08.022.

71. V. Bheemreddy, B. J. J. Liu, A. Wills, and C. P. Murcia, "Life Prediction Model Development for Flexible Photovoltaic Modules Using Accelerated Damp Heat Testing," IEEE 7th World Conf. on Photovoltaic Energy Conversion (WCPEC) 2018: 1249–1251.

72. T. Takamoto, H. Juso, K. Ueda, et al., "IMM Triple-Junction Solar Cells and Modules Optimized for Space and Terrestrial Conditions,"

Proceedings of the 44th IEEE Photovoltaic Specialist Conference (PVSC) (2017), https://doi.org/10.1109/PVSC.2017.8366097.

73. https://www.pv-magazine.com/2025/03/06/utmolight-pushes-perovksite-pv-module-efficiency-to-18-1/#:~:text=UtmoLight%20has%20reached%2018.1%25%20efficiency,Renewable%20Energy%20Laboratory%20(NREL).

74. A. Slade and V. Garboushian, "27.6% Efficient Silicon Concentrator Cell for Mass Production," Technical Digest, 15th International Photovoltaic Science and Engineering Conference, Shanghai (2005).

75. J. S. Ward, K. Ramanathan, F. S. Hasoon, et al., "A 21.5% Efficient Cu(In,Ga)Se_2 Thin-Film Concentrator Solar Cell," *Progress in Photovoltaics: Research and Applications* 10 (2002): 41–46.

76. F. Dimroth, T. N. D. Tibbits, M. Niemeyer, et al., "Four-Junction Wafer-Bonded Concentrator Solar Cells," *IEEE Journal of Photovoltaics* 6, no. 1 (2016): 343–349, https://doi.org/10.1109/JPHOTOV.2015.2501729.

77. NREL Press Release NR-4514, 2014.

78. "Press Release, Sharp Corporation," 2012, http://sharp-world.com/corporate/news/120531.html.

79. N. Jain, K. L. Schulte, J. F. Geisz, et al., "High-Efficiency Inverted Metamorphic 1.7/1.1 eV GaInAsP/GaInAs Dual-Junction Solar Cells," *Applied Physics Letters* 112 (2018): 053905.

80. M. Steiner, G. Siefer, T. Schmidt, M. Wiesenfarth, F. Dimroth, and A. W. Bett, "43% Sunlight to Electricity Conversion Efficiency Using CPV," *IEEE Journal of Photovoltaics* 6, no. 4 (2016): 1020–1024, https://doi.org/10.1109/JPHOTOV.2016.2551460.

81. M. A. Green, M. J. Keevers, I. Thomas, J. B. Lasich, K. Emery, and R. R. King, "40% Efficient Sunlight to Electricity Conversion," *Progress in Photovoltaics: Research and Applications* 23, no. 6 (2015): 685–691.

82. C. J. Chiang and E. H. Richards. "A 20% Efficient Photovoltaic Concentrator Module," Conf. Record, 21st IEEE Photovoltaic Specialists Conference, Kissimimee, 1990: 861–863.

83. http://amonix.com/pressreleases/amonix-achieves-world-record-359-module-efficiency-rating-nrel-4, accessed October 23, 2013.

84. S. van Riesen, M. Neubauer, A. Boos, et al., "New Module Design With 4-Junction Solar Cells for High Efficiencies," in Proceedings of the 11th Conference on Concentrator Photovoltaic Systems, 2015.

85. J. F. Martínez, M. Steiner, M. Wiesenfarth, G. Siefer, S. W. Glunz, and F. Dimroth, "Power Rating Procedure of Hybrid CPV/PV Bifacial Modules," *Progress in Photovoltaics: Research and Applications* 29 (2021): 614–629.

86. F. Zhang, S. R. Wenham, and M. A. Green, "Large Area, Concentrator Buried Contact Solar Cells," *IEEE Transactions on Electron Devices* 42 (1995): 144–149.

87. L. H. Slooff, E. E. Bende, A. R. Burgers, et al., "A Luminescent Solar Concentrator With 7.1% Power Conversion Efficiency," *Physica Status Solidi (RRL)* 2, no. 6 (2008): 257–259.

88. M. Steiner, M. Wiesenfarth, J. F. Martínez, G. Siefer, and F. Dimroth, "Pushing Energy Yield With Concentrating Photovoltaics," *AIP Conference Proceedings* 2149 (2019): 060006, https://doi.org/10.1063/1.5124199.

89. H. Müllejans, S. Winter, M. A. Green, E. D. Dunlop, "What Is the Correct Efficiency for Terrestrial Concentrator PV Devices? 38th European Photovoltaic Solar Energy Conference," 2021.

90. C. A. Gueymard, D. Myers, and K. Emery, "Proposed Reference Irradiance Spectra for Solar Energy Systems Testing," *Solar Energy* 73 (2002): 443–467.

91. "Program Milestones and Decision Points for Single Junction Thin Films," Annual Progress Report 1984, Photovoltaics, Solar Energy Research Institute, Report DOE/CE-0128, 1985.

92. I. Sakata, Y. Tanaka, and K. Koizawa, "Japan's New National R&D Program for Photovoltaics," Photovoltaic Energy Conversion, Conference Record of the 2006 IEEE 4th World Conference, 1, 2008, 1–4.

93. A. Jäger-Waldau. "PVNET: European Roadmap for PV R&D, EUR 21087 EN," 2004.

94. U. Rau, B. Blank, T. C. M. Müller, and T. Kirchartz, "Efficiency Potential of Photovoltaic Materials and Devices Unveiled by Detailed-Balance Analysis," *Physical Review Applied* 7 (2017): 044016.